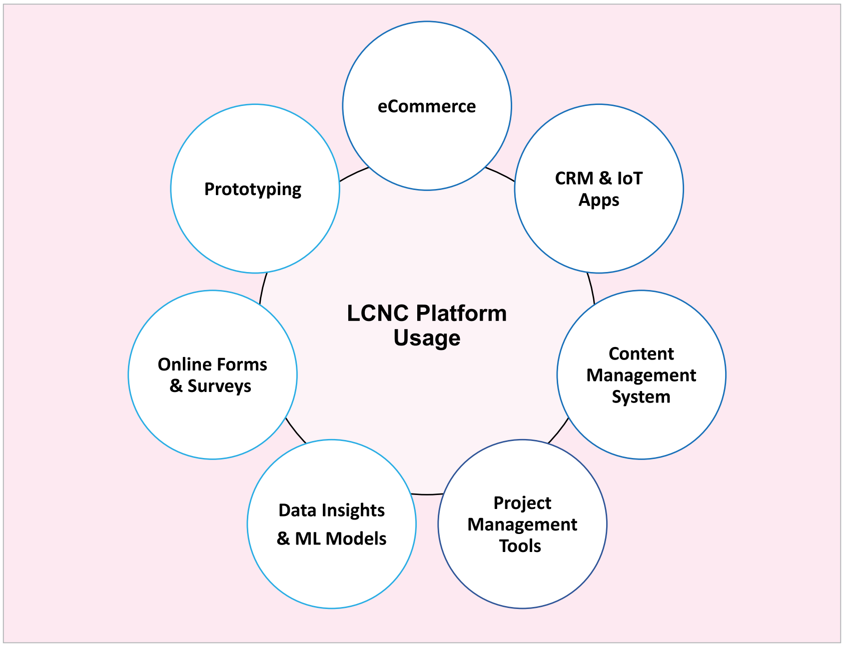

Figure 2: LCNC platform usage

Open source LCNC platforms

Open source LCNC platforms offer a range of capabilities ranging from web development and process automation to machine learning, making them valuable for developers and non-developers. They are cost efficient, flexible, scalable, reliable, transparent and, importantly, have no vendor lock-in.

The top ten open source LCNC development platforms are listed below.

Budibase: This versatile low-code platform offers a drag-and-drop interface, pre-built components, and support for multiple external data sources, including MongoDB, PostgreSQL, MySQL, Elasticsearch, Airtable, and more.

Huginn: Designed for process automation, it allows users to create agents that monitor automation of tasks and workflows without extensive coding. Huginn's agents create and consume events visualised via a directed graph.

WordPress: Widely used for building websites with a drag-and-drop interface, WordPress is highly customisable with numerous plugins and themes, making it a versatile no-code platform for web development.

Node-RED: This platform provides a visual programming interface to wire together devices, APIs, and online services, making it easy to create complex workflows. It offers a good solution for building IoT applications, and has nodes dedicated to Raspberry Pi and BeagleBone boards.

PyCaret: This low-code machine learning library in Python simplifies the process of building and deploying machine learning models, making it accessible for users with minimal coding experience

Baserow: This platform offers a drag-and-drop interface and data visualisation tools used for data-driven projects. It allows data collaboration, supports rich field types, APIs, webhooks and relations between tables.

Appsmith: Used for building internal tools, it provides a drag-and-drop interface and integrates with various data sources to create custom applications.

Directus: This CMS and data platform allows managing content and data with a no-code interface. It is highly customisable, integrates well with other tools and allows building workflow automations. It connects to PostgreSQL, MySQL, SQLite, OracleDB, CockroachDB, MariaDB, and MS SQL databases.

Supabase: This platform provides a backend as a service, including a real-time database, user authentication, row level security, REST API, edge functions and storage. All the data is accessible through a no-code interface.

Metabase: This business intelligence tool allows creation of dashboards and visualisations without writing code. It is user-friendly and integrates with various data sources.

Any application can be built based on the flexibility and extensibility of an LCNC platform. Some of these platforms are suitable for specific use cases such as, enterprise applications, e-commerce systems, IoT solutions or game development.

Challenges faced in the adoption of LCNC platforms

Vendor lock-in: Vendor lock-in is a challenge as many LCNC platforms do not allow moving the code or product from one platform to another.

Security: Safety of enterprise data is a major concern and is dependent on the platform being used.

Lack of customisation: Some LCNC platforms limit customisation possibilities.

Rigid templates and components: LCNC platforms provide readily available templates and components. Users need to modify these to meet a set of use cases while building an app.

Limited integration options: LCNC platforms focus on operational efficiency and do not allow custom integrations for third-party or internal enterprise systems.

No support for complex apps: Current LCNC platforms do not support complex functionality and unstructured workflows.